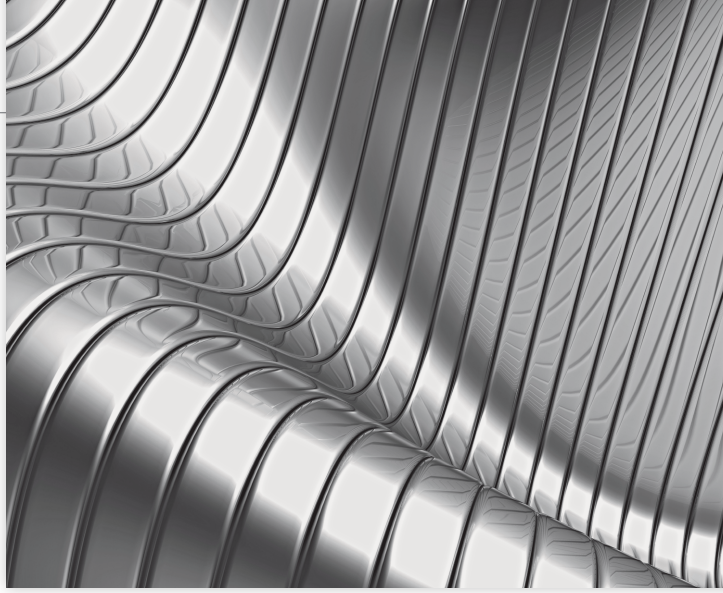


Preface

Thomas' Calculus, Fourteenth Edition in SI Units, provides a modern introduction to calculus that focuses on developing conceptual understanding of the underlying mathematical ideas. This text supports a calculus sequence typically taken by students in STEM fields over several semesters. Intuitive and precise explanations, thoughtfully chosen examples, superior figures, and time-tested exercise sets are the foundation of this text. We continue to improve this text in keeping with shifts in both the preparation and the goals of today's students, and in the applications of calculus to a changing world.

Many of today's students have been exposed to calculus in high school. For some, this translates into a successful experience with calculus in college. For others, however, the result is an overconfidence in their computational abilities coupled with underlying gaps in algebra and trigonometry mastery, as well as poor conceptual understanding. In this text, we seek to meet the needs of the increasingly varied population in the calculus sequence. We have taken care to provide enough review material (in the text and appendices), detailed solutions, and a variety of examples and exercises, to support a complete understanding of calculus for students at varying levels. Within the text, we present the material in a way that supports the development of mathematical maturity, going beyond memorizing formulas and routine procedures, and we show students how to generalize key concepts once they are introduced. References are made throughout, tying new concepts to related ones that were studied earlier. After studying calculus from *Thomas*, students will have developed problem-solving and reasoning abilities that will serve them well in many important aspects of their lives. Mastering this beautiful and creative subject, with its many practical applications across so many fields, is its own reward. But the real gifts of studying calculus are acquiring the ability to think logically and precisely; understanding what is defined, what is assumed, and what is deduced; and learning how to generalize conceptually. We intend this book to encourage and support those goals.

New to This Edition

We welcome to this edition a new coauthor, Christopher Heil from the Georgia Institute of Technology. He has been involved in teaching calculus, linear algebra, analysis, and abstract algebra at Georgia Tech since 1993. He is an experienced author and served as a consultant on the previous edition of this text. His research is in harmonic analysis, including time-frequency analysis, wavelets, and operator theory.

This is a substantial revision. Every word, symbol, and figure was revisited to ensure clarity, consistency, and conciseness. Additionally, we made the following text-wide updates:

- Updated graphics to bring out clear visualization and mathematical correctness.
- Added examples (in response to user feedback) to overcome conceptual obstacles. See Example 3 in Section 16.1.
- Added new types of homework exercises throughout, including many with a geometric nature. The new exercises are not just more of the same, but rather give different perspectives on and approaches to each topic. We also analyzed aggregated student usage and performance data from MyLab Math for the previous edition of this text. The results of this analysis helped improve the quality and quantity of the exercises.
- Added short URLs to historical links that allow students to navigate directly to online information.
- Added new marginal notes throughout to guide the reader through the process of problem solution and to emphasize that each step in a mathematical argument is rigorously justified.

New to MyLab Math®

Many improvements have been made to the overall functionality of MyLab Math since the previous edition. Beyond that, we have also increased and improved the content specific to this text.

- Instructors now have more exercises than ever to choose from in assigning homework.
- The MyLab Math exercise-scoring engine has been updated to allow for more robust coverage of certain topics, including differential equations.
- A full suite of Interactive Figures have been added to support teaching and learning. The figures are designed to be used in lecture, as well as by students independently. The figures are editable using the freely available GeoGebra software. The figures were created by Marc Renault (Shippensburg University), Kevin Hopkins (Southwest Baptist University), Steve Phelps (University of Cincinnati), and Tim Brzezinski (Berlin High School, CT).
- Enhanced Sample Assignments include just-in-time prerequisite review, help keep skills fresh with distributed practice of key concepts (based on research by Jeff Hieb of University of Louisville), and provide opportunities to work exercises without learning aids (to help students develop confidence in their ability to solve problems independently).
- Additional Conceptual Questions augment text exercises to focus on deeper, theoretical understanding of the key concepts in calculus. These questions were written by faculty at Cornell University under an NSF grant. They are also assignable through Learning Catalytics.
- Setup & Solve exercises now appear in many sections. These exercises require students to show how they set up a problem as well as the solution, better mirroring what is required of students on tests.
- New instructional videos by Greg Wisloski and Dan Radelet (both of Indiana University of PA) augment the already robust collection within the course. These videos support the overall approach of the text—specifically, they go beyond routine procedures to show students how to generalize and connect key concepts.

Content Enhancements

Chapter 1

- Shortened 1.4 to focus on issues arising in use of mathematical software and potential pitfalls. Removed peripheral material on regression, along with associated exercises.
- Added new Exercises: **1.1:** 59–62, **1.2:** 21–22; **1.3:** 64–65, **PE:** 29–32.

Chapter 2

- Added definition of average speed in 2.1.
- Clarified definition of limits to allow for arbitrary domains. The definition of limits is now consistent with the definition in multivariable domains later in the text and with more general mathematical usage.
- Reworded limit and continuity definitions to remove implication symbols and improve comprehension.
- Added new Example 7 in 2.4 to illustrate limits of ratios of trig functions.
- Rewrote 2.5 Example 11 to solve the equation by finding a zero, consistent with previous discussion.
- Added new Exercises: **2.1:** 15–18; **2.2:** 3h–k, 4f–i; **2.4:** 19–20, 45–46; **2.6:** 69–72; **PE:** 49–50; **AAE:** 33.

Chapter 3

- Clarified relation of slope and rate of change.
- Added new Figure 3.9 using the square root function to illustrate vertical tangent lines.
- Added figure of $x \sin(1/x)$ in 3.2 to illustrate how oscillation can lead to nonexistence of a derivative of a continuous function.
- Revised product rule to make order of factors consistent throughout text, including later dot product and cross product formulas.
- Added new Exercises: **3.2:** 36, 43–44; **3.3:** 51–52; **3.5:** 43–44, 61bc; **3.6:** 65–66, 97–99; **3.7:** 25–26; **3.8:** 47; **AAE:** 24–25.

Chapter 4

- Added summary to 4.1.
- Added new Example 3 with new Figure 4.27 to give basic and advanced examples of concavity.
- Added new Exercises: **4.1:** 61–62; **4.3:** 61–62; **4.4:** 49–50, 99–104; **4.5:** 37–40; **4.6:** 7–8; **4.7:** 93–96; **PE:** 1–10; **AAE:** 19–20, 33. Moved Exercises 4.1: 53–68 to PE.

Chapter 5

- Improved discussion in 5.4 and added new Figure 5.18 to illustrate the Mean Value Theorem.
- Added new Exercises: **5.2:** 33–36; **PE:** 45–46.

Chapter 6

- Clarified cylindrical shell method.
- Added introductory discussion of mass distribution along a line, with figure, in 6.6.
- Added new Exercises: **6.1:** 15–16; **6.2:** 45–46; **6.5:** 1–2; **6.6:** 1–6, 19–20; **PE:** 17–18, 35–36.

Chapter 7

- Added explanation for the terminology "indeterminate form."
- Clarified discussion of separable differential equations in 7.4.
- Replaced $\sin^{-1}$ notation for the inverse sine function with arcsin as default notation in 7.6, and similarly for other trig functions.
- Added new Exercises: **7.2:** 5–6, 75–76; **7.3:** 5–6, 31–32, 123–128, 149–150; **7.6:** 43–46, 95–96; **AAE:** 9–10, 23.

Chapter 8

- Updated 8.2 Integration by Parts discussion to emphasize $u(x)v'(x)\,dx$ form rather than $u\,dv$. Rewrote Examples 1–3 accordingly.
- Removed discussion of tabular integration and associated exercises.
- Updated discussion in 8.5 on how to find constants in Partial Fraction method.
- Updated notation in 8.8 to align with standard usage in statistics.
- Added new Exercises: **8.1:** 41–44; **8.2:** 53–56, 72–73; **8.3:** 75–76; **8.4:** 49–52; **8.5:** 51–66, 73–74; **8.8:** 35–38, 77–78; **PE:** 69–88.

Chapter 9

- Clarified the different meaning of a sequence and a series.
- Added new Figure 9.9 to illustrate sum of a series as area of a histogram.
- Added to 9.3 a discussion on the importance of bounding errors in approximations.

- Added new Figure 9.13 illustrating how to use integrals to bound remainder terms of partial sums.
- Rewrote Theorem 10 in 9.4 to bring out similarity to the integral comparison test.
- Added new Figure 9.16 to illustrate the differing behaviors of the harmonic and alternating harmonic series.
- Renamed the nth term test the "nth term test for divergence" to emphasize that it says nothing about convergence.
- Added new Figure 9.19 to illustrate polynomials converging to $\ln(1 + x)$, which illustrates convergence on the half-open interval $(-1, 1]$.
- Used red dots and intervals to indicate intervals and points where divergence occurs and blue to indicate convergence throughout Chapter 9.
- Added new Figure 9.21 to show the six different possibilities for an interval of convergence.
- Added new Exercises: **9.1:** 27–30, 72–77; **9.2:** 19–22, 73–76, 105; **9.3:** 11–12, 39–42; **9.4:** 55–56; **9.5:** 45–46, 65–66; **9.6:** 57–82; **9.7:** 61–65; **9.8:** 23–24, 39–40; **9.9:** 11–12, 37–38; **PE:** 41–44, 97–102.

Chapter 10

- Added new Example 1 and Figure 10.2 in 10.1 to give a straightforward first example of a parametrized curve.
- Updated area formulas for polar coordinates to include conditions for positive r and non-overlapping θ.
- Added new Example 3 and Figure 10.37 in 10.4 to illustrate intersections of polar curves.
- Added new Exercises: **10.1:** 19–28; **10.2:** 49–50; **10.4:** 21–24.

Chapter 11

- Added new Figure 11.13(b) to show the effect of scaling a vector.
- Added new Example 7 and Figure 11.26 in 11.3 to illustrate projection of a vector.
- Added discussion on general quadric surfaces in 11.6, with new Example 4 and new Figure 11.48 illustrating the description of an ellipsoid not centered at the origin via completing the square.
- Added new Exercises: **11.1:** 31–34, 59–60, 73–76; **11.2:** 43–44; **11.3:** 17–18; **11.4:** 51–57; **11.5:** 49–52.

Chapter 12

- Added sidebars on how to pronounce Greek letters such as kappa, tau, etc.
- Added new Exercises: **12.1:** 1–4, 27–36; **12.2:** 15–16, 19–20; **12.4:** 27–28; **12.6:** 1–2.

Chapter 13

- Elaborated on discussion of open and closed regions in 13.1.
- Standardized notation for evaluating partial derivatives, gradients, and directional derivatives at a point, throughout the chapter.
- Renamed "branch diagrams" as "dependency diagrams" which clarifies that they capture dependence of variables.
- Added new Exercises: **13.2:** 51–54; **13.3:** 51–54, 59–60, 71–74, 103–104; **13.4:** 20–30, 43–46, 57–58; **13.5:** 41–44; **13.6:** 9–10, 61; **13.7:** 61–62.

Chapter 14

- Added new Figure 14.21b to illustrate setting up limits of a double integral.
- Added new 14.5 Example 1, modified Examples 2 and 3, and added new Figures 14.31, 14.32, and 14.33 to give basic examples of setting up limits of integration for a triple integral.
- Added new material on joint probability distributions as an application of multivariable integration.
- Added new Examples 5, 6 and 7 to Section 14.6.
- Added new Exercises: **14.1:** 15–16, 27–28; **14.6:** 39–44; **14.7:** 1–22.

Chapter 15

- Added new Figure 15.4 to illustrate a line integral of a function.
- Added new Figure 15.17 to illustrate a gradient field.
- Added new Figure 15.19 to illustrate a line integral of a vector field.
- Clarified notation for line integrals in 15.2.
- Added discussion of the sign of potential energy in 15.3.
- Rewrote solution of Example 3 in 15.4 to clarify connection to Green's Theorem.
- Updated discussion of surface orientation in 15.6 along with Figure 15.52.
- Added new Exercises: **15.2:** 37–38, 41–46; **15.4:** 1–6; **15.6:** 49–50; **15.7:** 1–6; **15.8:** 1–4.

Chapter 16

- Added new Example 3 with Figure 16.3 to illustrate how to construct a slope field.
- Added new Exercises: **16.1:** 11–14; **PE:** 17–22, 43–44.

Appendices: Rewrote Appendix 7 on complex numbers.

Continuing Features

Rigor The level of rigor is consistent with that of earlier editions. We continue to distinguish between formal and informal discussions and to point out their differences. Starting with a more intuitive, less formal approach helps students understand a new or difficult concept so they can then appreciate its full mathematical precision and outcomes. We pay attention to defining ideas carefully and to proving theorems appropriate for calculus students, while mentioning deeper or subtler issues they would study in a more advanced course. Our organization and distinctions between informal and formal discussions give the instructor a degree of flexibility in the amount and depth of coverage of the various topics. For example, while we do not prove the Intermediate Value Theorem or the Extreme Value Theorem for continuous functions on a closed finite interval, we do state these theorems precisely, illustrate their meanings in numerous examples, and use them to prove other important results. Furthermore, for those instructors who desire greater depth of coverage, in Appendix 6 we discuss the reliance of these theorems on the completeness of the real numbers.

Writing Exercises Writing exercises placed throughout the text ask students to explore and explain a variety of calculus concepts and applications. In addition, the end of each chapter contains a list of questions for students to review and summarize what they have learned. Many of these exercises make good writing assignments.

End-of-Chapter Reviews and Projects In addition to problems appearing after each section, each chapter culminates with review questions, practice exercises covering the entire chapter, and a series of Additional and Advanced Exercises with more challenging or synthesizing problems. Most chapters also include descriptions of several **Technology Application Projects** that can be worked by individual students or groups of students over a longer period of time. These projects require the use of *Mathematica* or *Maple*, along with pre-made files that are available for download within MyLab Math.

Writing and Applications This text continues to be easy to read, conversational, and mathematically rich. Each new topic is motivated by clear, easy-to-understand examples and is then reinforced by its application to real-world problems of immediate interest to students. A hallmark of this book has been the application of calculus to science and engineering. These applied problems have been updated, improved, and extended continually over the last several editions.

Technology In a course using the text, technology can be incorporated according to the taste of the instructor. Each section contains exercises requiring the use of technology; these are marked with a T if suitable for calculator or computer use, or they are labeled **Computer Explorations** if a computer algebra system (CAS, such as *Maple* or *Mathematica*) is required.

Additional Resources

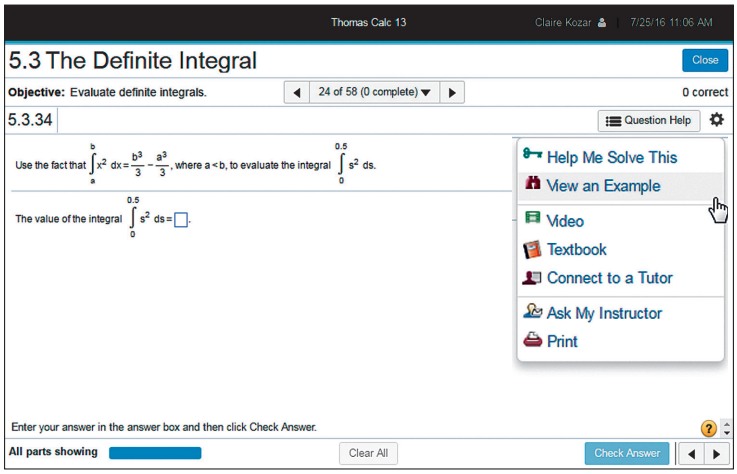
MyLab Math® Online Course (access code required)

Built around Pearson's best-selling content, MyLab Math is an online homework, tutorial, and assessment program designed to work with this text to engage students and improve results. MyLab Math can be successfully implemented in any classroom environment—lab-based, hybrid, fully online, or traditional.

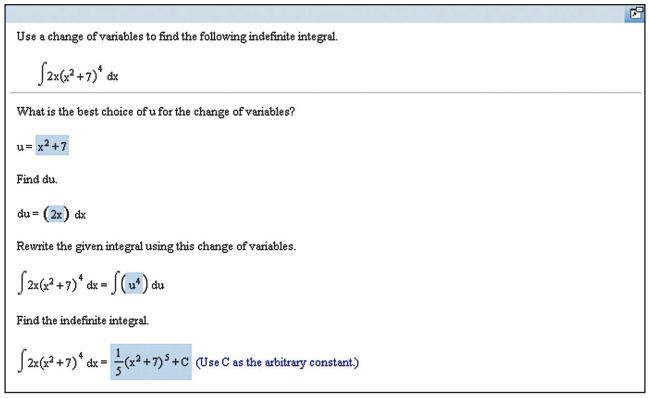
Used by more than 37 million students worldwide, MyLab Math delivers consistent, measurable gains in student learning outcomes, retention, and subsequent course success. Visit **www.mymathlab.com/results** to learn more.

Motivation Students are motivated to succeed when they're engaged in the learning experience and understand the relevance and power of mathematics. MyLab Math online homework offers students immediate feedback and tutorial assistance that motivates them to do more, which means they retain more knowledge and improve their test scores.

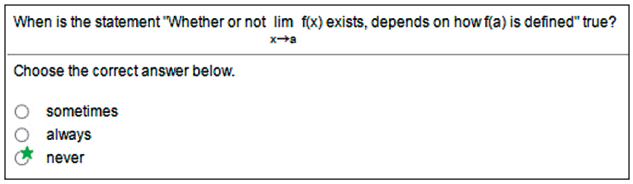
- **Exercises with immediate feedback**—the assignable exercises for this text regenerate algorithmically to give students unlimited opportunity for practice and mastery. MyLab Math provides helpful feedback when students enter incorrect answers and includes optional learning aids such as Help Me Solve This, View an Example, videos, and an eText.



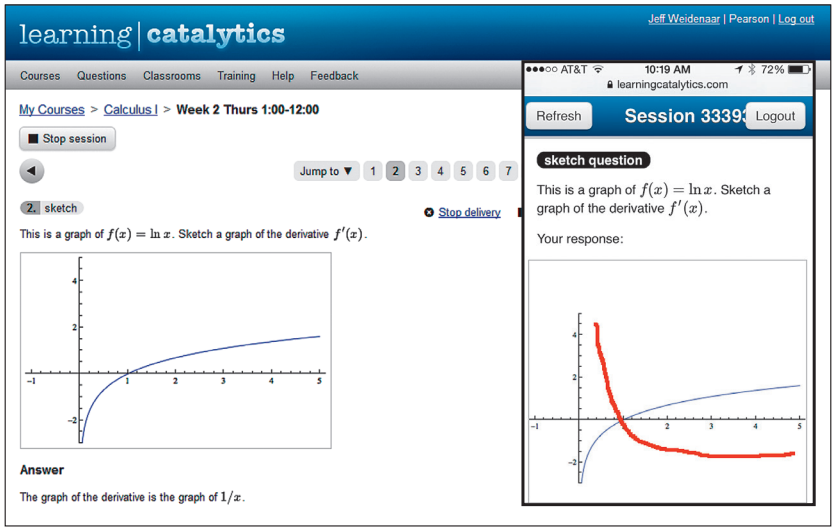
- **Setup and Solve Exercises** ask students to first describe how they will set up and approach the problem. This reinforces students' conceptual understanding of the process they are applying and promotes long-term retention of the skill.



- **Additional Conceptual Questions** focus on deeper, theoretical understanding of the key concepts in calculus. These questions were written by faculty at Cornell University under an NSF grant and are also assignable through Learning Catalytics.

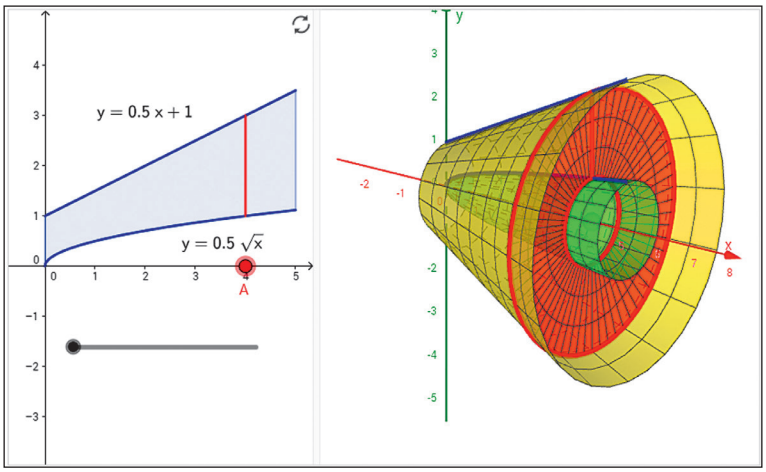


- **Learning Catalytics™** is a student response tool that uses students' smartphones, tablets, or laptops to engage them in more interactive tasks and thinking during lecture. Learning Catalytics fosters student engagement and peer-to-peer learning with real-time analytics. Learning Catalytics is available to all MyLab Math users.



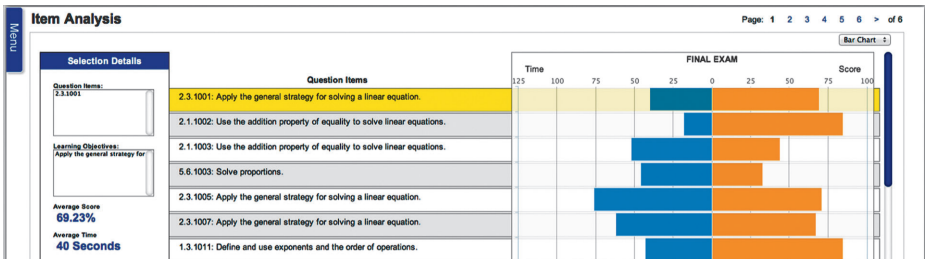
Learning and Teaching Tools

- **Interactive Figures** illustrate key concepts and allow manipulation for use as teaching and learning tools. We also include videos that use the Interactive Figures to explain key concepts.



- **Instructional videos**—hundreds of videos are available as learning aids within exercises and for self-study. The Guide to Video-Based Assignments makes it easy to assign videos for homework by showing which MyLab Math exercises correspond to each video.
- **The complete eText** is available to students through their MyLab Math courses.

- **PowerPoint Presentations** that cover each section of the book are available for download.
- **Mathematica manual and projects, Maple manual and projects, TI Graphing Calculator manual**—These manuals cover *Maple* 17, *Mathematica* 8, and the TI-84 Plus and TI-89, respectively. Each provides detailed guidance for integrating the software package or graphing calculator throughout the course, including syntax and commands.
- **Accessibility** and achievement go hand in hand. MyLab Math is compatible with the JAWS screen reader, and it enables students to read and interact with multiple-choice and free-response problem types via keyboard controls and math notation input. MyLab Math also works with screen enlargers, including ZoomText, MAGic, and SuperNova. And, all MyLab Math videos have closed-captioning. More information is available at **http://mymathlab.com/accessibility**.
- **A comprehensive gradebook** with enhanced reporting functionality allows you to efficiently manage your course.
  - **The Reporting Dashboard** offers insight as you view, analyze, and report learning outcomes. Student performance data is presented at the class, section, and program levels in an accessible, visual manner so you'll have the information you need to keep your students on track.



  - **Item Analysis** tracks class-wide understanding of particular exercises so you can refine your class lectures or adjust the course/department syllabus. Just-in-time teaching has never been easier!

MyLab Math comes from an experienced partner with educational expertise and an eye on the future. Whether you are just getting started with MyLab Math, or have a question along the way, we're here to help you learn about our technologies and how to incorporate them into your course. To learn more about how MyLab Math helps students succeed, visit **www.mymathlab.com** or contact your Pearson rep.

Instructor's Solutions Manual (downloadable)

ISBN: 1-29-225326-6 | 978-1-29225326-8

The Instructor's Solutions Manual contains complete worked-out solutions to all the exercises in *Thomas' Calculus*. It can be downloaded from within MyLab Math or the Pearson Instructor Resource Center, **www.pearsonglobaleditions.com/irc**.

Student's Solutions Manual

ISBN: 1-29-225331-2 | 978-1-29-225331-2

The Student's Solutions Manual contains worked-out solutions to all the odd-numbered exercises in *Thomas' Calculus*. These manuals can be downloaded from within MyLab Math.

Companion Website

The companion Website, located at **www.pearsonglobaleditions.com**, includes opportunities for practice and review.

Just-In-Time Algebra and Trigonometry for Calculus, Fourth Edition

ISBN: 0-321-67104-X | 978-0-321-67104-2

Sharp algebra and trigonometry skills are critical to mastering calculus, and *Just-in-Time Algebra and Trigonometry for Calculus* by Guntram Mueller and Ronald I. Brent is designed to bolster these skills while students study calculus. As students make their way through calculus, this brief supplementary text is with them every step of the way, showing them the necessary algebra or trigonometry topics and pointing out potential problem spots. The easy-to-use table of contents has topics arranged in the order in which students will need them as they study calculus. This supplement is available in printed form only (note that MyLab Math contains a separate diagnostic and remediation system for gaps in algebra and trigonometry skills).

Technology Manuals and Projects (downloadable)

Maple Manual and Projects by Marie Vanisko, Carroll College
Mathematica Manual and Projects by Marie Vanisko, Carroll College
TI-Graphing Calculator Manual by Elaine McDonald-Newman, Sonoma State University

These manuals and projects cover Maple 17, Mathematica 9, and the TI-84 Plus and TI-89. Each manual provides detailed guidance for integrating a specific software package or graphing calculator throughout the course, including syntax and commands. The projects include instructions and ready-made application files for Maple and Mathematica. These materials are available to download within MyLab Math.

TestGen®

ISBN: 1-29-225332-0 | 978-1-29-225332-9

TestGen® (**www.pearsoned.com/testgen**) enables instructors to build, edit, print, and administer tests using a computerized bank of questions developed to cover all the objectives of the text. TestGen is algorithmically based, allowing instructors to create multiple but equivalent versions of the same question or test with the click of a button. Instructors can also modify test bank questions or add new questions. The software and test bank are available for download from Pearson Education's online catalog, **www.pearsonglobaleditions.com**.

PowerPoint® Lecture Slides

ISBN: 1-29-225330-4 | 978-1-29-225330-5

These classroom presentation slides were created for the *Thomas' Calculus* series. Key graphics from the book are included to help bring the concepts alive in the classroom. These files are available to qualified instructors through the Pearson Instructor Resource Center, **www.pearsonglobaleditions.com/irc**, and within MyLab Math.

Acknowledgments

We are grateful to Duane Kouba, who created many of the new exercises. We would also like to express our thanks to the people who made many valuable contributions to this edition as it developed through its various stages:

Accuracy Checkers

Thomas Wegleitner
Jennifer Blue
Lisa Collette

Reviewers for the Fourteenth Edition

Alessandro Arsie, *University of Toledo*
Doug Baldwin, *SUNY Geneseo*
Steven Heilman, *UCLA*
David Horntrop, *New Jersey Institute of Technology*
Eric B. Kahn, *Bloomsburg University*
Colleen Kirk, *California Polytechnic State University*
Mark McConnell, *Princeton University*
Niels Martin Møller, *Princeton University*
James G. O'Brien, *Wentworth Institute of Technology*
Alan Saleski, *Loyola University Chicago*
Alan Von Hermann, *Santa Clara University*
Don Gayan Wilathgamuwa, *Montana State University*
James Wilson, *Iowa State University*

The following faculty members provided direction on the development of the MyLab Math course for this edition.

Charles Obare, *Texas State Technical College, Harlingen*
Elmira Yakutova-Lorentz, *Eastern Florida State College*
C. Sohn, *SUNY Geneseo*
Ksenia Owens, *Napa Valley College*
Ruth Mortha, *Malcolm X College*
George Reuter, *SUNY Geneseo*
Daniel E. Osborne, *Florida A&M University*
Luis Rodriguez, *Miami Dade College*
Abbas Meigooni, *Lincoln Land Community College*
Nader Yassin, *Del Mar College*
Arthur J. Rosenthal, *Salem State University*
Valerie Bouagnon, *DePaul University*
Brooke P. Quinlan, *Hillsborough Community College*
Shuvra Gupta, *Iowa State University*
Alexander Casti, *Farleigh Dickinson University*
Sharda K. Gudehithlu, *Wilbur Wright College*
Deanna Robinson, *McLennan Community College*
Kai Chuang, *Central Arizona College*
Vandana Srivastava, *Pitt Community College*
Brian Albright, *Concordia University*
Brian Hayes, *Triton College*
Gabriel Cuarenta, *Merced College*
John Beyers, *University of Maryland University College*
Daniel Pellegrini, *Triton College*
Debra Johnsen, *Orangeburg Calhoun Technical College*
Olga Tsukernik, *Rochester Institute of Technology*
Jorge Sarmiento, *County College of Morris*
Val Mohanakumar, *Hillsborough Community College*
MK Panahi, *El Centro College*
Sabrina Ripp, *Tulsa Community College*
Mona Panchal, *East Los Angeles College*
Gail Illich, *McLennan Community College*
Mark Farag, *Farleigh Dickinson University*
Selena Mohan, *Cumberland County College*

Global Edition

The publishers would like to thank the following for their contribution to the Global Edition:

Contributor for the Fourteenth Edition in SI Units

José Luis Zuleta Estrugo received his PhD degree in Mathematical Physics from the University of Geneva, Switzerland. He is currently a faculty member in the Department of Mathematics in École Polytechnique Fédérale de Lausanne (EPFL), Switzerland, where he teaches undergraduate courses in linear algebra, calculus, and real analysis.

Reviewers for the Fourteenth Edition in SI Units

Fedor Duzhin, *Nanyang Technological University*
B. R. Shankar, *National Institute of Technology Karnataka*

Contributor for the Thirteenth Edition in SI Units

Antonio Behn, *Universidad de Chile*

Dedication

We regret that prior to the writing of this edition our coauthor Maurice Weir passed away. Maury was dedicated to achieving the highest possible standards in the presentation of mathematics. He insisted on clarity, rigor, and readability. Maury was a role model to his students, his colleagues, and his coauthors. He was very proud of his daughters, Maia Coyle and Renee Waina, and of his grandsons, Matthew Ryan and Andrew Dean Waina. He will be greatly missed.